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Assessment TOOL-1

CO1 - AT1.

1. Scenario: Event communication security

(a) Encrypt the message using a shift of 4 positions.

The caesar cipher shifts each letter forward by four positions in the alphabet.

Original message: MEET AT NOON

Plaintext:- M E E T A T .N O O N

Shift +4:- Q I I X E X R S S R

Encrypted message: QIIX EX RSSR.

b. Decryption Process:-

An authorized number shifts each encrypted letter backward by four positions to recover the original message.

Example:-

Q I X E

| | | |

M E T A.

∴ The decrypted message is meet At NOON.

2. Scenario : Intercepted suspicious message.

(a) Deduce the original message:

The intercepted message is GSRH RH Z HVXIVG.

use the Atbash, in which A maps to Z, B maps to Y and so, we obtain:

- G SRH - THIS
- RH - IS
- Z - A
- HVXIVG - secret.

original message : This is a secret.

b. Justification:

The Atbash cipher replaces every letter with its opposite letter in the alphabet.

Examples:-

- G - T
- S - H
- R - I
- H - S.

The repeating word structures and the recognizable English phrase support this interpretation. The Atbash cipher uses a fixed substitution rule, making it easy to identify and decode without a secret key.

3. Scenario: Secure Messaging System:

(a) Encrypt Hello using the keyword KEY.

The Vigenere cipher uses a repeating keyword to determine the shift applied to each plaintext letter.

Plaintext : Hello.

Repeated keyword : KEYKE.

using A = 0 , B = 1 ... Z = 25 encryption is performed

using $C = P + K \bmod 26$

plain text:	H	E	L	L	O
key word :	K	E	Y	K	E
shift value:	10	4	24	10	4
ciphertext:	R	I	J	V	S

Encrypted message: RIJVS.

(b) Influence of the repeating keyword:-

The keyword key repeats until it matches the length of the plaintext. Each keyword letter determines the shift applied to its corresponding plaintext letter.

For example

H + K = R

E + E = I

L + Y = J

The same plaintext letter can produce different ciphertext letters when paired with different keyword letters, However, a short repeating keyword is vulnerable to cryptanalysis. So this method is not suitable for modern secure communications.

4. Scenario: Suspicious digital Asset Investigation:

Investigators can use the following techniques to detect steganography, which is the practice of hiding information inside an ordinary file.

1. visual inspection: Examine the image for unusual colors, distortion or pixel patterns.

2. Statical analysis:- Analyze pixel values and color channel distributions for irregularities.

3. Metadata Examination: Inspect file properties, metadata and unexpected embedded information.

4. File analysis tools: Use forensic tools such as Steganopse or Stegsolve to identify possible hidden content.

5. File size comparition: Compare the image's size and structure with an original or similar image to identify suspicious differences.

b. Two techniques to strengthen security:-

1. Encrypt the hidden data before Embedding:

Use a strong encryption algorithm such as AES-256 so that extracted information remains unreadable without the decryption.

2. Use robust steganographic methods:

Select embedding techniques that minimize statistical changes and distribute the hidden information carefully across suitable image components.

5. Scenario : Legay encryption in corporate communication.

(a) The ciphertext is:-

WECRLTEERDSOEEFEAOCAVDEN.

This resembles a three-rail Rail Fence decryption to the supplied ciphertext does not produce a clear English sentence. Therefore, the cipher text may use a different transposition arrangement or may contain an error.

For reference, the commonly used three-rail Fence example

Plaintext:- WEAREDISCOVERED FLEEAT ONCE.

Cipher :- WECRLTEERDSSOEEFEAOCAIVDEN.

(b) why is this method vulnerable ?

- It does not replace letters, only rearranges ~~can be tested~~ on their positions.
- The original letter frequencies remain unchanged.
- A small no. of possible rail arrangements can be tested.
- It does not provide strong protection against modern computational

(c) Practical enchancment:-

Replace the legacy transortion cipher with AES-256 encryption using a secure mode such AES-GCM. This protects the confidentiality of the message.